



CODECLASH - QUANTDEVS

Brand Style Guide

This serves as a definitive visual language guide, defining every design decision, for CodeClash - a competitive coding and mathematics gaming platform.

v3.0

September 2026

WCAG 2.2 AA

Tailwind v4

MVVM Architecture

MISSION/GOAL

To make learning programming and mathematics engaging through real-time competitive challenges that inspire continuous skill development and problem-solving.

TARGET AUDIENCE

Students, aspiring programmers, and mathematic enthusiasts who want to improve their coding and problem-solving skills through engaging, competitive learning.

PERSONALITY

By default: bold, and cosmic where the UI feels stepping into a mission control centre.

AESTHETIC

A black cosmic background with maroon accents, pink gradients, clean typography and light theme support.

01 - COLOR PALETTE

Color System



Brand Style Guide

v3.0



Background

#0a0008

RGB 10,0,8, HSL 312,100,2

Default page background, sidebar base

WCAG AAA with Text Primary (17.9:1)

Primary

#c0395a

RGB 192,57,90, HSL 345,54,49

Buttons, sidebar active highlight, brand accent --primary

WCAG AA with White (5.3:1)

Primary Dark

#530a23

RGB 83,10,35, HSL 339,78,18

Depth accents, gradients, --primary-dark

WCAG AAA with Text Primary (12.7:1)

Secondary

#FFEFE0

RGB 255,239,224, HSL 29,100,94

Soft card surfaces, light backgrounds

WCAG AAA with Secondary Text (12.7:1)

Button Primary

#C0395A

RGB 192,57,90, HSL 345,54,49

Primary buttons

WCAG AA on #FFFFFF (4.6:1)

Button Secondary

#FFEFE0

RGB 255,239,224, HSL 29,100,94

Secondary buttons

WCAG AAA on #530A24 (10.2:1)

Button Primary Text

#FFFFFF

RGB 255,255,255, HSL 0,0,100

Primary button text

WCAG AA on Button Primary (5.3:1)

Button Secondary Text

#fcecdd

RGB 252,236,221, HSL 29,84,93

Secondary button text, link accents

WCAG AAA on Background (17.9:1)

Text Primary

#FCECDD

RGB 252,236,221, HSL 29,84,93

Primary text on all dark backgrounds

WCAG AAA on Background (17.9:1)

Text Secondary

#530A24

RGB 83,10,36, HSL 339,78,18

Text on light/secondary surfaces

WCAG AAA on Secondary (12.7:1)



Brand Style Guide

v3.0

#E93577

RGB 233,53,119, HSL 338,80,56

HOver state for primary buttons

WCAG AA on White (4.0:1)

#4CAF50

RGB 76,175,80, HSL 122,39,49

Success states, confirmations, positive feedback

WCAG AA on dark (4.5:1)

Danger

#E53935

RGB 229,57,53, HSL 1,77,55

Destructive actions, errors, validation fails

WCAG AA on white (4.5:1)

Warning

#e8a33d

RGB 232,163,61, HSL 36,79,57

Warning states, medium difficulty, pending status

WCAG AAA on itself (7.8:1)

Info

#3d9be8

RGB 61,155,232, HSL 207,79,57

Information

WCAG AA on itself (6.0:1)

PINK COLORS - FULL RAMP



#ED5A90

#E93577

#DC1860

#B91551

#931040

#6E0C30

#530A24

#490820

#250410

02 - TYPOGRAPHY

Type System

The sole typeface is Roboto - chosen for its readability, weight range, and geometric clarity. The font is sourced from Google Fonts under the SIL Open Font License (OFL) v1.1 and the Apache License 2.0, completely free to use for both personal and commercial purposes.



Brand Style Guide

v3.0

HEADING BIG 5rem/w700 --heading-big-size

CodeClash Gaming

Display titles - .heading-big class

HEADING 3rem/w700 --heading-size

Welcome Back, Challenger

Page headings, section titles - .heading class

HEADING SUB 1.9rem/w400 --font-size-md

Build your skills. Earn your rank.

Subheadings, taglines, back buttons - .heading-sub class

BODY-LARGE 2.3rem/w500 --font-size-l

Large body text, prominent labels

BODY-MEDIUM 1.9rem/w500 --font-size-md

General body text, paragraphs

SMALL 1.3rem/w400 --font-size-sm

Already have an account?

Captions, helper texts, form labels - .fields class

EXTRA SMALL 1rem/w400 --font-size-xsm

Select a game mode and start competing

Statistics, small UI labels - text-xsm

SCORE DISPLAY 1.9rem/w400 --font-dseg

2026

Digital, game type numerals for ELO, streaks, scores - .score-display

FONT FAMILY

'Roboto', sans-serif

SOURCE

Google Fonts

LICENCE



and the Apache
License 2.0

WEIGHTS

400, 500, 600, 700,
800

LINE HEIGHT

1.6 - 1.7

LETTER SPACING

0.01rem (headings)

03 - LOGO AND ICONOGRAPHY

Logo System

The CodeClash wordmark, icon systems, and asset guidelines. UI icons use Lucide React and thematic assets use custom PNGs.

LOGO VARIANTS**CodeClash**

Background
Main (default) usage

CodeClash

Primary - Dark
Dark surfaces

CodeClash

Primary - Light
Light surfaces

PERMITTED

- Scale proportionally maintaining aspect ratio
- Use on approved dark #530A24 or light #FFEF00 backgrounds

FORBIDDEN

- Stretch or distort the logo in any dimension
- Visual effects such as shadows, glows or outlines
- Color outside of the approved palette
- Place on clashing background
- Rotate, skew or transform



Brand Style Guide

v3.0

All UI icons use **Lucide React** - outline style, consistent 2px stroke weight. Install via `npm install lucide-react`

20px / w-5 h-5

Header icons - Search, Bot, UserCircle

24px / w-6 h-6

Standalone / decorative icons



Search



UserCircle



Bot



Dashboard



Help Menu



Trophy



BarChart



Medal



Users



Settings

- Always use outline style - never filled

- Stroke weight: 2px default

- Import individually: `import { Search } from "lucide-react";`

- Color via Tailwind text

- If using filled, never mix filled and outline in the same context

THEMATIC ASSETS - CUSTOM PNGS

Thematic assets that define the robots in space identity. These are not interchangeable with the UI icons. Located in `src/assets/`

assets/Robots/**HelloRobot_Pink, Pink_fighting, Pink_celebrate, arms_up**

Welcome page mascot, dashboard avatar, match screens

assets/Planets/**Earth**

SignIn page - bottom beam source

assets/Background/**dashboard, SignInBeam, SignUpBeam, SymbolBackground**

Page backgrounds and light beam effects

assets/Avatar/**blue_avatar, placeholder, purple_avatar**

The match screen avatars

assets/Decor/**door, RedUFO**

Thematic decor

w-[480px]**w-full h-full object-cover****w-[220px]**



backgrounds

- Use object-contain to prevent distortion
- Do not upscale PNGs beyond their natural resolution
- Do not change the transformations or scales of source images
- Use object-cover for full page backgrounds
- Do not use thematic assets as UI icons

04 - DESIGN TOKENS

Token System

All tokens are CSs custom properties defined in global.css. The @theme inline block maps them to Tailwind utilities. Click any row to copy the token reference.

COLOR TOKENS

TOKEN	VALUE	USAGE
--background	#0a0008	Default page background, sidebar base
--background-elevated	rgba(252, 236, 221, 0.03)	Elevated surface overlay
--background-card	rgba(252, 236, 221, 0.05)	Card surfaces
--primary	#c0395a	Brand accent, sidebar active highlight
--primary-dark	#530a23	Depth accents, gradients
--secondary	#fcecdd	Secondary buttons, soft card surfaces
--primary-text	#fcecdd	Primary text on all dark backgrounds
--secondary-text	#530a24	Text on light/secondary surfaces
--button-primary	#c0395a	Primary buttons
--button-secondary	transparent	Secondary buttons
--button-text-primary	ffffff	Text on primary buttons
--button-text-secondary	#fcecdd	Text on secondary buttons
--accent	#e93577	Hover state for primary buttons
--accent-text	#fcecdd	Text on accent surfaces
--success	#4caf50	Success states, confirmations, positive feedback
--danger	#e53935	Destructive actions, errors, validation fails
--warning	#e9a33d	Warning states, medium difficulty, pending status
--info	#3d9be8	Information
--difficulty-easy/medium/hard	success.warning/danger	Question difficulty badges
--life-primary	#c0395a	Life bar
--text	#fcecdd	White text fallback
--muted	rgba(252, 236, 221, 0.5)	Muted surface overlay



	<code>0.18)</code>	
<code>--input</code>	<code>rgba(252, 236, 221, 0.06)</code>	Form input background
<code>--ring</code>	<code>#c0395a</code>	Focus ring color

RADIUS TOKENS

TOKEN	VALUE	USAGE
<code>--radius-sm</code>	<code>calc(var(--radius) - 4px)</code>	16px - small elements
<code>--radius-md</code>	<code>calc(var(--radius) - 2px)</code>	18px - medium elements
<code>--radius-lg</code>	<code>var(--radius)</code>	20px - standard inputs and buttons
<code>--radius-xl</code>	<code>calc(var(--radius) + 4px)</code>	24px - large elements

TYPOGRAPHY TOKENS

TOKEN	VALUE	USAGE
<code>--font</code>	<code>'Roboto', sans-serif</code>	Primary font - body, headings
<code>--font-logo</code>	<code>'Baloo Bhai 2', sans-serif</code>	Logo display
<code>--heading</code>	<code>'Roboto', sans-serif</code>	Heading font
<code>--heading-weight</code>	<code>700</code>	Bold - .heading and .heading-big
<code>--font-weight</code>	<code>500</code>	Medium - default body font
<code>--heading-sub-weight</code>	<code>400</code>	Regular - .heading-sub
<code>--heading-size</code>	<code>3rem</code>	Standard heading - .heading
<code>--heading-big-size</code>	<code>5rem</code>	Display heading - .heading-big
<code>--font-size-xsm</code>	<code>1rem</code>	Extra small - text-xsm
<code>--font-size-sm</code>	<code>1.3rem</code>	Small - .fields, captions
<code>--font-size-md</code>	<code>1.9rem</code>	Medium - .heading-sub, body text
<code>--font-size-l</code>	<code>2.3rem</code>	Large - prominent body text
<code>--font-size-xl</code>	<code>3rem</code>	Extra large
<code>--font-size-2xl</code>	<code>3.3rem</code>	2X large
<code>--font-size-3xl</code>	<code>3.6rem</code>	3X large
<code>--font-dseg</code>	<code>DSEG7, monospace</code>	Digital, game type numerals for ELO, streaks, scores

SHADOW TOKENS

TOKEN	VALUE	USAGE
<code>--badge-shadow</code>	<code>0 4px 6px rgba(0, 0, 0, 0.3)</code>	Buttons, badges - .badge-shadow
<code>--card-shadow</code>	<code>0rem 0.2rem 0.5rem rgba(0, 0, 0, 0.25)</code>	Cards and elevated surfaces - .card-shadow
<code>--glow-shadow</code>	<code>0 0 24px rgba(192, 57, 90, 0.45)</code>	Animated glow effects - .animate-glow

05 - COMPONENT LIBRARY

Components



Button Primary

Button Secondary

Back Button

Input Field

Checkbox

Glass Card

Progress Bar

Badges

Score Display

Sign up

Hover

Disabled

Loading..

VARIANTS

Default

Hover

Active

Disabled

Loading

CLASSES & TOKENS

bg-button-primary

text-button-text-primary

rounded-lg

shadow-badge

font-bold

NOTES

w-[100%] on auth pages, fixed width on dashboard. h-[3rem], font-size 1.5rem, hover: -translate-y-px

06 - LAYOUT & SPACING

Grid & Spacing

The system uses distinct layout patterns. All share the same spacing scale and token system.

SIDEBAR NAVIGATION

The sidebar is implemented via shadcn's `SidebarProvider` and `AppSidebar` component. It wraps all authenticated routes via the `Layout` component in `layout.tsx`



Leaderboard
/leaderboard

Badges
/badges

Friends
/friends

SPACING SCALE

xs		0.25rem	4px	Icon gaps
sm		0.5rem	8px	Tight padding
md		1rem	16px	Component padding
lg		1.5rem	24px	Card padding, section gaps
xl		2rem	32px	Page padding, large gaps
xxl		3rem	48px	Section margins

07 - ACCESSIBILITY

Accessibility

Conformance target is WCAG 2.2 minimum. AAA is achieved for all body text paitings on the primary dark background.

CONFORMANCE
WCAG 2.2 AA

BODY TEXT CONTRAST
17.9:1

BUTTON CONTRAST
5.3:1

THEME SUPPORT
Dark(default) and Light

CONSTRAST PAIRS

FOREGROUND	BACKGROUND	RATIO	LEVEL



Brand Style Guide

v3.0

 Secondary Text	 Secondary	12.7:1	AAA
 Button Text Secondary	 Secondary	4.8:1	AA
 White	 Success	4.5:1	AA
 White	 Danger	4.5:1	AA

ACCESSIBILITY RULES

- Conformance target: WCAG 2.2 AA minimum, AAA achieved for all body text pairings
- All form inputs have visible labels or aria-label attributes
- Buttons use type="button" explicitly to prevent unintended form submissions
- Logical reading order top-to-bottom, left-to-right

KEYBOARD NAVIGATION

- Buttons activate using Enter or Space
- No keyboard traps exist
- Dismissal may be done using Escape

SCREEN READER SUPPORT

- Every form field has an associated label
- Decorative images use empty alt attributes
- Meaningful icons include accessible labels

REDUCED MOTION

- Animations are decorative only
- Users can navigate without animation

THEME SUPPORT



Decorative elements like banners adapt to active theme

08 - VOICE & TONE

Writing Style

CodeClash is direct, and motivating. The platform speaks with confidence - never pleading, nor vague.

Direct

No filler. Say it as it is.

Motivating

Challenges are opportunities to rise.

Confident

The platform knows what it is doing.

DO

"Sign up"

Short, and direct

"Already have an account?"

Conversational

"Password must be at least 8 characters"

Specific

"Code sent! Check your email"

Confirmation of action

DONT

"SIGN UP NOW"

Capital letters, and pushy

"Please enter your password"

Pleading

"Error"

Vague with no action item or path forward

"Password too short"

Vague and blame with no solution

09 - CHANGELOG

Demo Changelogs

What changed between Demos with rationale for each decision. This proves the guide has evolved based on real implementation decisions.



Demo 1 -> Demo 2

Colors

5 changes

- New color palette - pinkColors documented as named Tailwind tokens
- Fixed --button-text-primary from #FFFF to #FFFFFF
- Defined --success #4CAF50 and --danger #E53935
- Added --muted and --muted-text for overlays and faded text usage
- Removed grey fallback, now --background will always resolve to maroon

Rationale: Several color tokens were invalid or undefined, causing inconsistencies and unwanted fallbacks. The entire color palette was also changed.

Typography

5 changes

- --heading corrected from Baloo Bhai 2 to Roboto
- --font-logo kept as Baloo Bhai 2 for logo display only
- --heading-big-size added and defined as 5rem
- --font-size-xsm added as 1rem
- --badge-font-size corrected from 0.1rem (invisible) to 0.75rem

Rationale: Demo 1 had undefined variables and mismatches. All the tokens now match what the actual component uses.

Styling Architecture

5 changes

- Migrated from per-file CSS to a single global.css with Tailwind v4
- All styling now done via Tailwind utility classes directly in TSX components
- global.css serves as the single source of truth for all design tokens via CSS custom properties
- @theme inline block maps all CSS variables to Tailwind utilities
- @layer base defines shared utility classes eg .heading, .heading-big, .heading-sub



Brand Style Guide

v3.0

global changes easier.

Architecture

4 changes

- Adopted MVVM - Model, View Model and View separation
- Validation logic extracted to pure functions in Models
- Auth state managed via AuthContext, and navigation via react-router-dom
- Dashboard uses Layout wrapper with AppSidebar via SidebarProvider

Rationale: Clean separation of concerns improves testability and makes changes easier, and safer.

Pages and Visual Design

4 changes

- Complete visual redesign from Demo 1
- New Robots in Space theme implemented across the App
- Dark maroon radial gradient established as the core, default background identity
- Theme related assets introduced such as UFO, planets, new robot mascot, symbol background texture, light beam effects

Rationale: Demo 1 screens had no visual identity, Demo 2 now fully establishes a cosmos theme that runs consistently across the App.

Demo 2 -> Demo 3

Colors

4 changes

- New color palette - documented in the new Brand Style Guide
- Added --warning and --info status colors alongside --success and --danger
- Added --difficulty-easy/medium/hard mirroring success/warning and danger for question status and difficulty
- Added a light mode palette via :root-light

Rationale: The new color palette separates background from brand accent and introduces a more layered look with consistent colors.



Brand Style Guide

v3.0

- --font-dseg for digital, game type numerals for ELO, streaks, scores

Rationale: For the app to have some kind of spacey/game feel and to make these stats and Elo ratings stand out.

Styling Architecture 3 changes

- Animated effects (.animate-glow)
- Card surfaces split into 3 variants - glass, elevated and glow
- Added .starfield and .confetti classes to reinforce the robots in space theme

Rationale: These were added to give the UI more depth, glow effects, motion.

Pages and Visual Design 2 changes

- Light theme introduced as an alternative look to the app
- A more black-ish cosmic theme with layered surfaces

Rationale: Demo 3 refines the robots in space theme with a more consistent language and introduces a light mode option for slight variety.

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